REPORT / DESIGN SCREENING

Generalized wheel-leg study

Jonathan V2 geometry + Andy sizing + independent suspension mechanics

A compact, non-2:1 candidate meets the central travel target. Its useful stroke is 285 mm after two geometric reserves. Spring and motor requirements depend on a separate supported-chassis fixture, actual mass allocation and motion time.

TOTAL CAPACITY

325 mm

includes both end reserves

USEFUL CAPACITY

285 mm

20 mm + 20 mm reserved

GUIDE RATIO

1.775

free variable; not fixed at 2

Conditional geometry

L2 = 180 mm; L3 = 340 mm; ra/rb = 106.5/60 mm; mounting alpha = 4.375 deg. Physical theta2 = 100 to 159.810888 deg; useful theta2 = 106.429341 to 157.077766 deg. Wheel radius = 200 mm. Summed links = 520 mm; full lateral excursion = 11.412 mm.

Result / fixture	Computed value
Fixed-A robust static score	46.260 Nm peak / 41.089 Nm time RMS
Supported chassis, 10 kg corner	Hip bellcrank, 1 s per direction, zero damping
Ideal joint motor demand, full cycle	6.449 Nm peak / 3.641 Nm RMS
Joint speed / acceleration	1.605 rad/s / 6.573 rad/s2
Held 4g TOTAL support, 11.25 kg	143.772 Nm joint-output stress envelope

Decision boundary

The geometry is a compact bounded-search example. Extending the tibia domain finds lower static torque at greater length and drift. No selected motor, manufactured spring, strength rating, tire-contact feasibility or 3D clearance has been established. Numerical agreement does not certify a canonical robot.

Report map: 2 reference; 3 coordinates; 4 constraints; 5-8 optimization; 9-10 springs; 11-12 motion; 13-14 impact envelope; 15 verification; 16 reproduction.

REFERENCE / ORIGINAL SOLVER

Screenshot case reproduced

Rebuilt from entered parameters; no private screenshot is retained in this report.

Input family	Original value
Links / wheel	L2 = L3 = 250 mm; rw = 200 mm radius (diameter 400 mm)
Guide radii / ratio	ra = 100; rb = 60 mm; qab = 5/3 = 1.666667
Drive radii	rc = rd = re = rf = 35 mm; qcd = qef = 1
Angles / current pose	te = 100; tc = 165; theta2 = 100.135 deg
Mounting / contact	alpha = 0; beta = 13.5 deg; phi0 = 0 assumed
Driving loads	Global Fy = Fz = +100 N; Fn/Ft derived
Masses / gravity	m1 = 10 kg; m2 = m3 = m5 = m6 = 0; g = 9.81 m/s2
Other explicit assumptions	r5I = 35 mm; Tab0 = Tcd0 = Tef0 = 0 N

At the captured pose

theta3 (deg)	tau2 (Nm)	tau4 (Nm)	Tight ab span (N)
53.24333	-25.62966	-14.77849	84.48598 / lower

Current C displacement is (0.265170, 0.337721) mm. Final displacement is (101.157122, 338.615931) mm. With 20 mm excluded at each end, useful vertical travel is 298.615931 mm. The lateral motion is a real property of this generalized geometry.

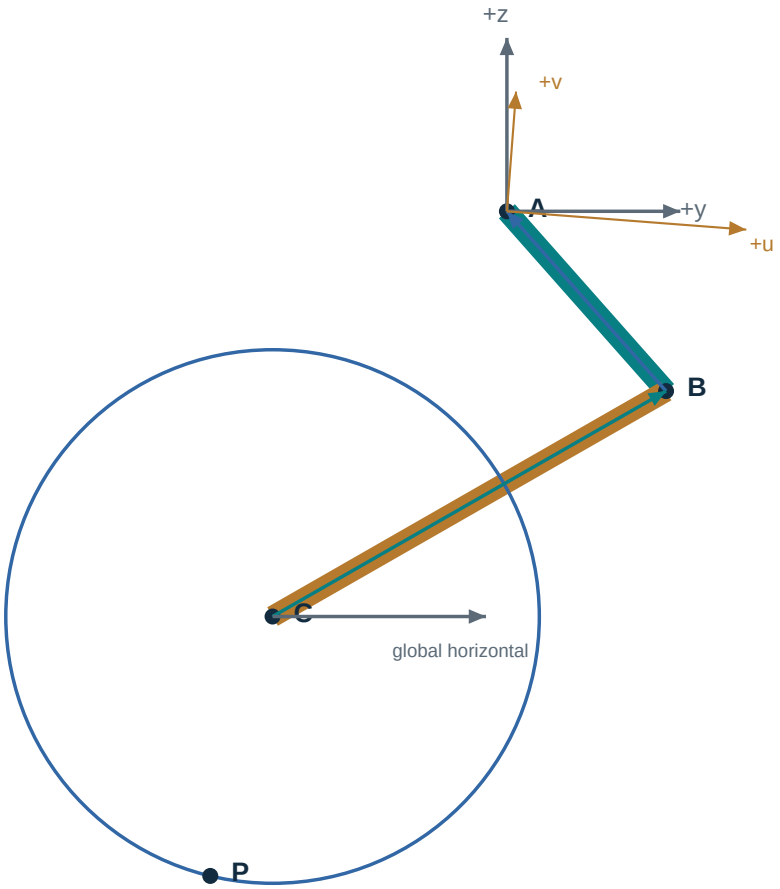
Original global maxima

Across theta2 = 100-165 and beta = 0-90 deg, alpha fixed at 0: |tau2| = 44.448948 Nm; |tau4| = 20 Nm; tight Tab = 337.983156 N. These are the original solver's attained analytical maxima, separate from this report's fixed-beta robust design objective.

Phi0 and preloads were not visible in the reference: zero is an explicit demonstration assumption. Massless moving links are an analytical limit, not an executable physical fixture.

One geometry, clear frames

A = hip; B = knee; C = wheel centre; P = contact patch. V2 names are preserved.



Angle directions

theta2 points B to A.
theta3 points C to B.
Both are CCW from local +u.
Alpha rotates u/v clockwise relative to y/z.
Beta is clockwise from -z toward -y.

Generalized linkage and drive

```
theta3 = phi0 + (180 - theta2)(qab - 1)
B = -L2 [cos(theta2), sin(theta2)]
C = B - L3 [cos(theta3), sin(theta3)]
dC/dq = dB/dq + (1-qab)L3 [sin(theta3),-cos(theta3)]
K = 1-qab + qef(qab-qcd); theta6 = K (theta2-te)
```

Actual wheel-centre upward angle

Measure the lower-link ray C to B against the **global** horizontal through C:
`angle = atan2(Bz-Cz, By-Cy) = wrapped(theta3-alpha)`
Central useful range: 52.642261 to 13.389732 deg. The expression 180-theta2 applies only to the equal-link, qab=2, phi0=alpha=0 special case.

Coordinate work conservation

Relative knee derivative is -qab. Therefore signed tau_knee = -tau2/qab, omega_knee = -qab*qdot. A different coordinate can lower torque while increasing speed; it does not improve ideal work or transmission efficiency.

REQUIREMENT / FEASIBILITY

Travel includes the reserves

Hard minimum useful stroke 200 mm; preferred about 250 mm; editable total 300-350 mm.

300 MM TOTAL

260 mm

usable; 20 mm reserved each end

325 MM TOTAL

285 mm

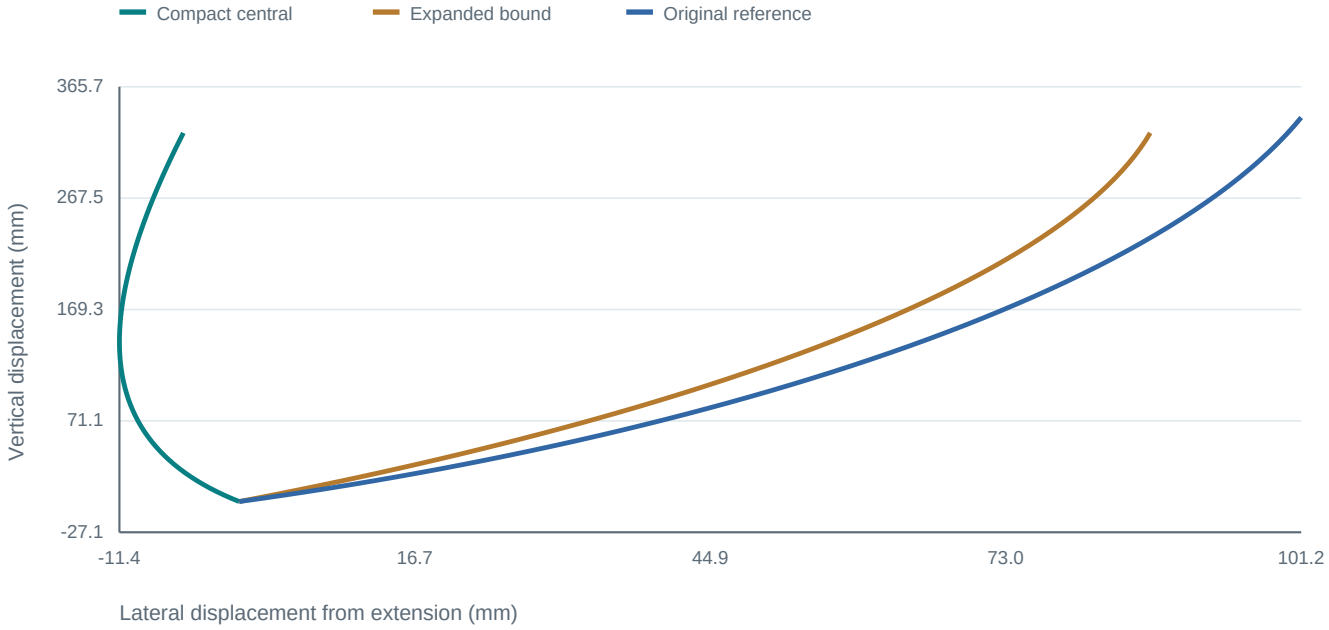
usable; 20 mm reserved each end

350 MM TOTAL

310 mm

usable; 20 mm reserved each end

Full physical wheel-centre paths



Constraint / assumption	Central candidate
Useful stops, 20 mm each	106.429341 to 157.077766 deg
Minimum useful vertical Jacobian	197.480 mm/rad (required >= 5)
Full lateral excursion	11.412 mm (limit 150)
Flat-ground chassis air gap	141.284 mm (depth 200, gap >= 20)
Projected hip-to-wheel rim gap	20.840 mm (assumed >= 20)

Wheel radius controls gap and contact moment, not automatic travel. The projected hip exclusion disk is a conservative optional assumption: staggering axial planes may resolve a projected overlap. The reference remains mathematically valid when that filter rejects it.

OPTIMIZATION / TWO METHODS

Bounded search, exact targets

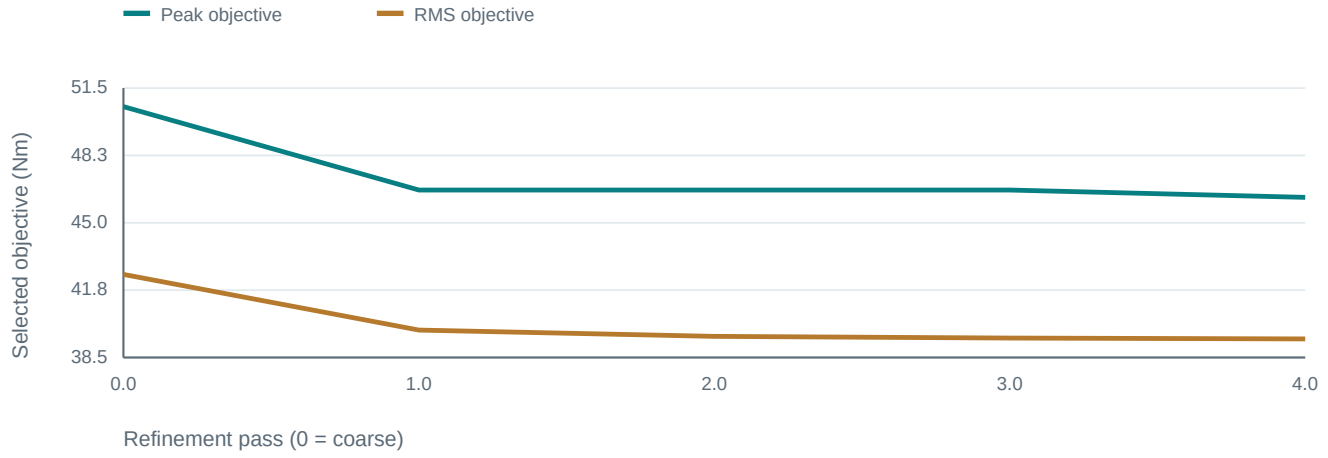
Loads and masses stay fixed; suspension/drive demand is scored in declared coordinates.

Scalar: coarse grid, feasible peak/RMS ranking, four leading seeds, halved coordinate steps. **Pareto:** torque/link sum/drift nondominance, then up to twelve frontier seeds. Both densely recheck finalists with at least 721 samples; exactTarget solves tc for the selected total.

Domains: L2 and L3 180-340 step40 mm; qab 1.4-2.4 step0.25; alpha 0-20 step10 deg; te 100-110 step10 deg; solved tc constrained to 155-170 deg. Off-grid upper endpoints included. 750 coarse geometries; four refinement passes for this six-row comparison.

Total/use mm	Score	L2/L3 mm	qab	alpha deg	Peak Nm	RMS Nm	Drift mm
300/260	peak	205.0/340	1.6500	0.000	44.627	38.368	33.82
300/260	rms	205.0/340	1.6656	0.000	44.720	38.337	28.38
325/285	peak	180.0/340	1.7750	4.375	46.260	41.089	11.41
325/285	rms	187.5/340	1.7750	7.500	49.029	39.443	21.80
350/310	peak	180.0/340	1.9000	6.875	49.980	45.158	42.08
350/310	rms	187.5/340	1.9000	12.500	54.528	42.016	20.36

Central-case convergence



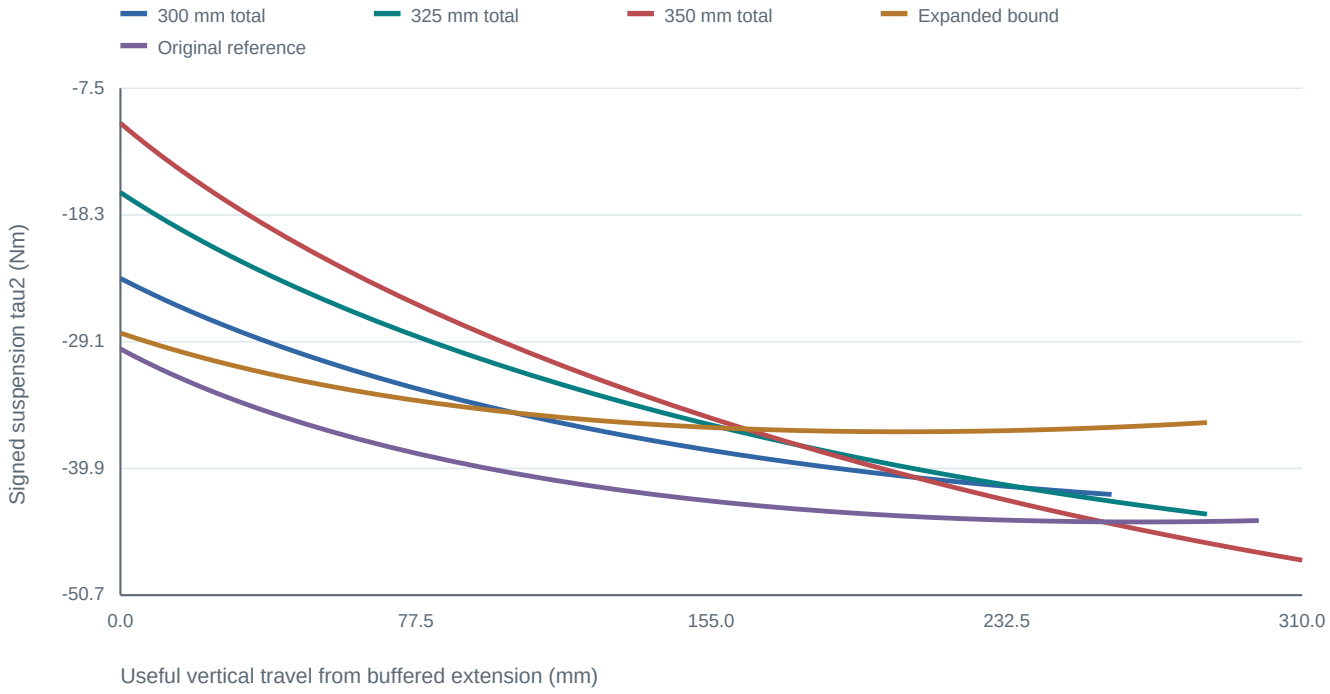
Robust score = worst case over Fy = -100, 0, +100 N at Fz = 100 N and beta = 13.5 deg. Peak/RMS use sqrt(tau2^2+tau4^2), an equal-weight two-actuator demand score, not one motor rating.

PROFILES / NOMINAL CONTACT

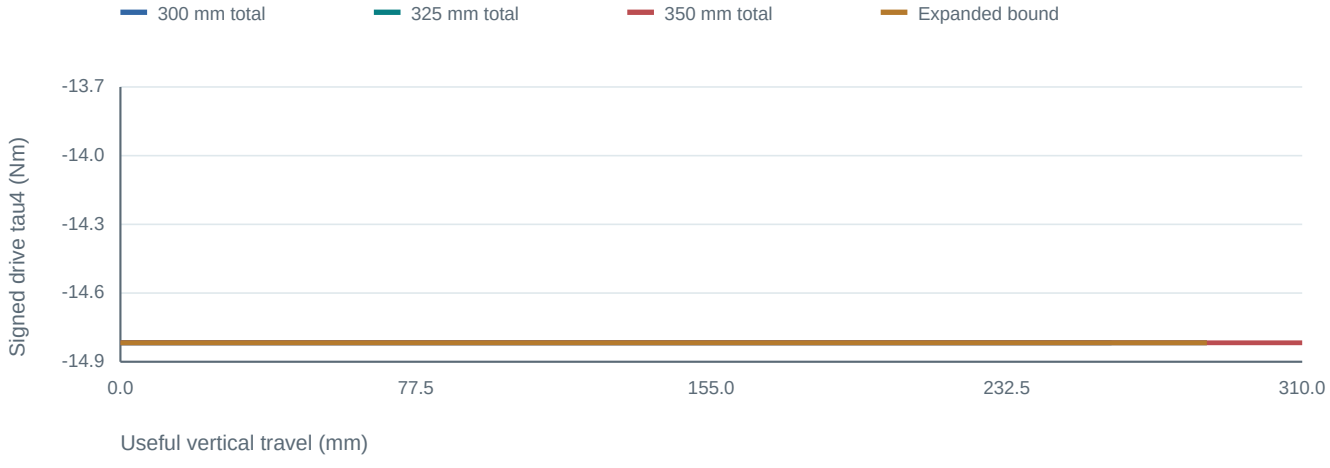
See the whole control window

Signed nominal traces retain horizontal-force assistance and actual drive torque.

Full suspension output profiles



Drive holding demand (qcd = qef = 1)



All four design cases have K=0 although qab varies. The nominal drive holding torque is -14.778491 Nm; the robust envelope reaches 24.116306 Nm magnitude under opposite Fy. These traces show the entered +100/+100 N force, while the search table uses the worst configured case.

Signed mean is not a sizing minimum. The central nominal |tau2| peak is 43.782204 Nm; combined nominal peak/RMS is 46.209146/37.041717 Nm. Ideal nominal motor work over useful compression is -28.291453 J; external prescribed-load work is +28.291460 J.

CAPABILITY / WORK AND TORQUE

Travel per torque demand

Compare the same loading, actuator coordinates and motion weighting.

Case / search	Useful stroke mm	Robust peak Nm	Time RMS Nm	Stroke/RMS mm per Nm	Actuator ROM deg
300 / peak	260	44.627	38.368	6.777	47.752
300 / rms	260	44.720	38.337	6.782	47.430
325 / peak	285	46.260	41.089	6.936	50.648
325 / rms	285	49.029	39.443	7.226	50.242
350 / peak	310	49.980	45.158	6.865	51.628
350 / rms	310	54.528	42.016	7.378	51.212

Peak, mean and RMS answer different questions

Peak is the greatest transient magnitude. **Mean absolute torque** describes typical demand under a stated weighting. **Signed mean** can cancel strong positive and negative loads and therefore cannot size a motor. **Time RMS** weights high demands quadratically and can inform copper-loss comparisons for one fixed motor/gearing/duty model. The combined sqrt(tau2^2+tau4^2) score here describes two output coordinates; it is not one motor's heating.

Torque-normalized capability is not efficiency

Central stroke/RMS = 6.936 mm/Nm. At fixed useful stroke, maximizing this is equivalent to minimizing the chosen RMS. It is dimensionful and does not account for angular ROM, speed, current, conversion losses or regeneration. Changing to a relative-knee coordinate changes torque and speed in opposite directions without increasing ideal work.

Ideal work and true efficiency

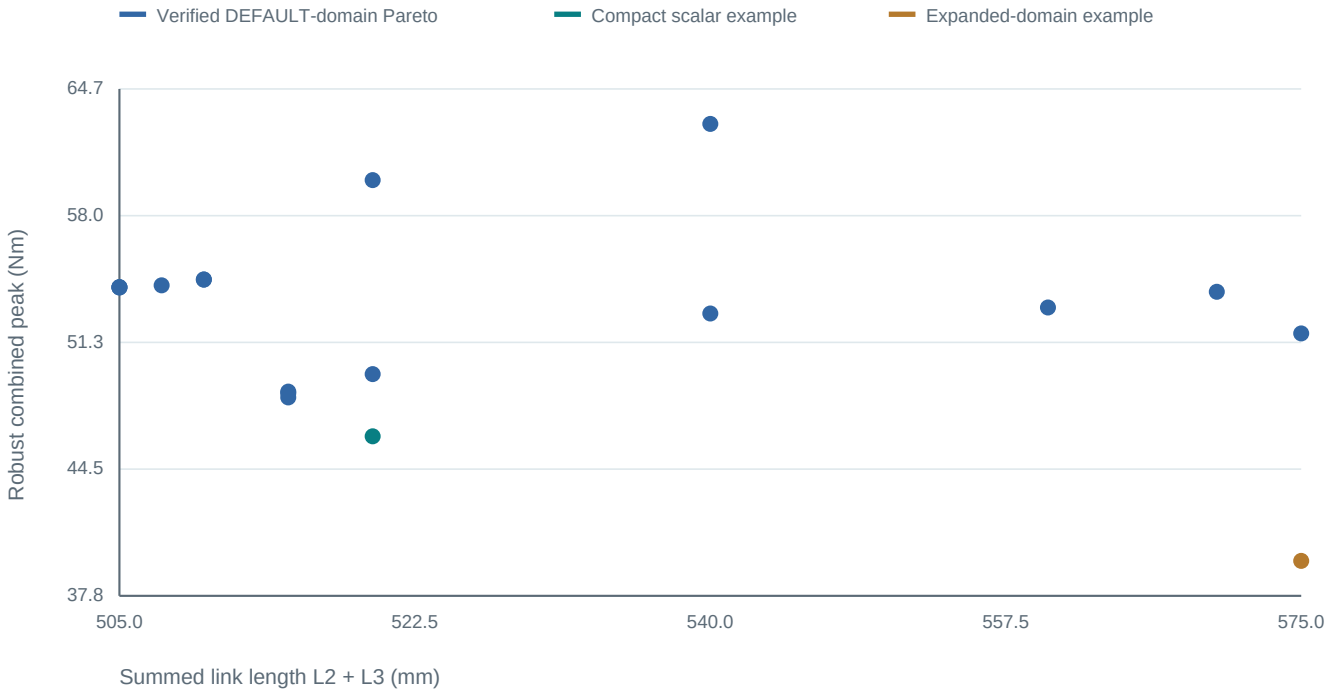
Ideal actuator work is W = integral(tau dq), not torque divided by height. External work includes Fy dy + Fz dz plus the wheel moment contribution Mw dtheta6. A travel-weighted mean alone hides actuator angular ROM. Physical efficiency would require useful work divided by electrical energy with measured losses, duty and regenerative behavior; those inputs are absent.

TRADEOFF / COMPACTNESS

Torque has a geometry cost

A boundary winner must be challenged before it can inform hardware selection.

Evaluated alternatives; dominance also uses full lateral excursion



325 mm total / 285 useful	Compact central	Expanded-bound diagnostic
L2 / L3; qab	180 / 340; 1.775	180 / 395; 1.400
alpha; tc (deg)	4.375; 159.810888	0; 169.525341
Robust peak / RMS Nm	46.260 / 41.089	39.632 / 37.180
Link sum / lateral drift mm	520 / 11.412	575 / 86.778
Ground / assumed hip gap mm	141.284 / 20.840	61.584 / 25.515

Expanding L3 to 500 mm (step40; 1,350 coarse geometries; three local passes) improves peak by 14.3%, but adds 55 mm link sum and 75.4 mm drift. L3=395 is interior; L2=180 and qab=1.4 still reach their lower bounds. Unknown moving masses, stiffness and strength can reverse this preference.

The default-domain Pareto run retained 16 nondominated verified alternatives after 824 evaluations. It compares the selected peak score, link sum and drift across the evaluated population. Neither method proves a continuum-wide or global hardware optimum.

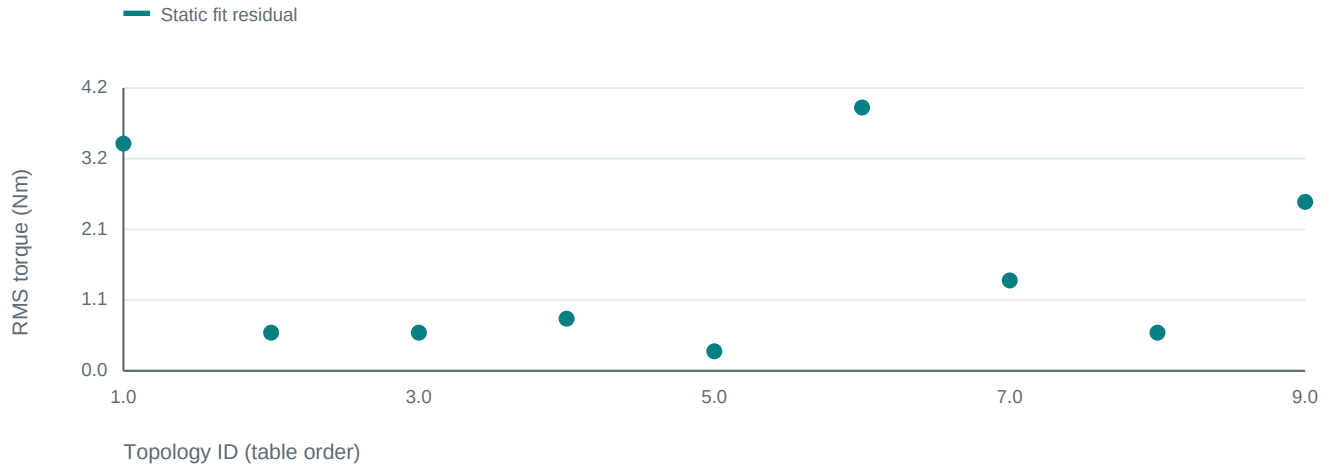
SPRINGS / CONDITIONAL FIT

Nine geometries, one load case

Supported hub C; chassis pitch fixed; physical masses total 10 kg; no damping.

Topology	Rate N/mm	Installed load N	Free mm	Stroke mm	Peak load N	Residual RMS Nm
1. Legacy tip	5.915	155.79	125.94	68.06	558.38	3.404
2. Hip pulley	3.325	229.23	659.96	70.72	464.38	0.612
3. Knee pulley	4.222	258.29	653.22	62.76	523.25	0.612
4. Direct scissor	0.296	184.44	862.63	121.10	220.22	0.819
5. Hip bellcrank	4.370	213.95	35.25	67.78	510.15	0.337
6. Knee bellcrank	8.108	128.42	53.04	63.12	640.22	3.937
7. Chassis direct	0.181	108.94	1,000.00	191.92	143.75	1.384
8. Knee capture	4.222	258.29	653.22	62.76	523.25	0.612
9. Gravity balance	3.109	213.19	200.00	73.14	440.58	2.543

Passive fit comparison



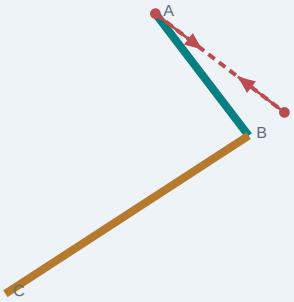
Rates fit 81 uniformly spaced suspension-angle samples, not the later motion-time RMS. All fit constraints are ideal: nonnegative rate, correct unilateral force signs and positive physical free/coil lengths. Chassis-direct free length reaches its 1,000 mm fit bound. Coil force ratings, solid height and packaging remain unverified.

SPRING FIGURES / FORCE SITES

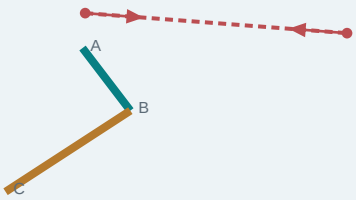
Applied forces at actual sites

Common useful midpoint pose; illustration uses the fixed-A translated frame.

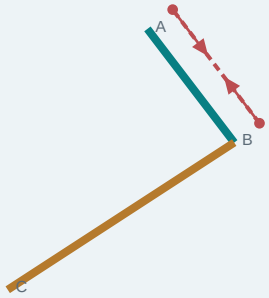
1. Legacy tip



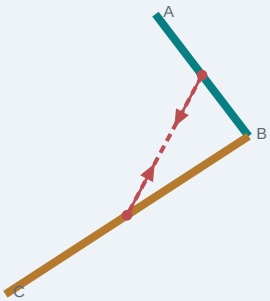
2. Hip pulley



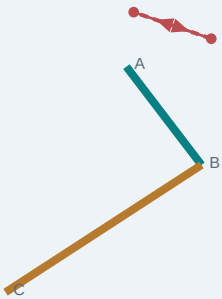
3. Knee pulley



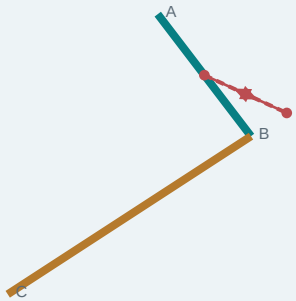
4. Direct scissor



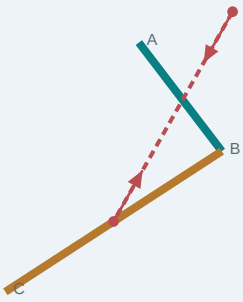
5. Hip bellcrank



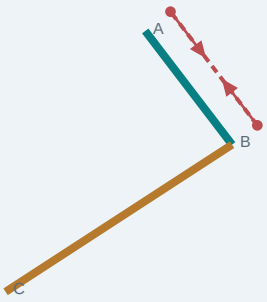
6. Knee bellcrank



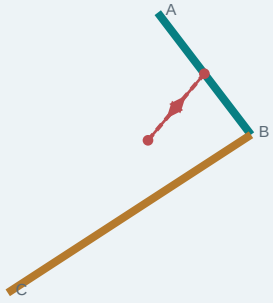
7. Chassis direct



8. Knee capture



9. Gravity balance



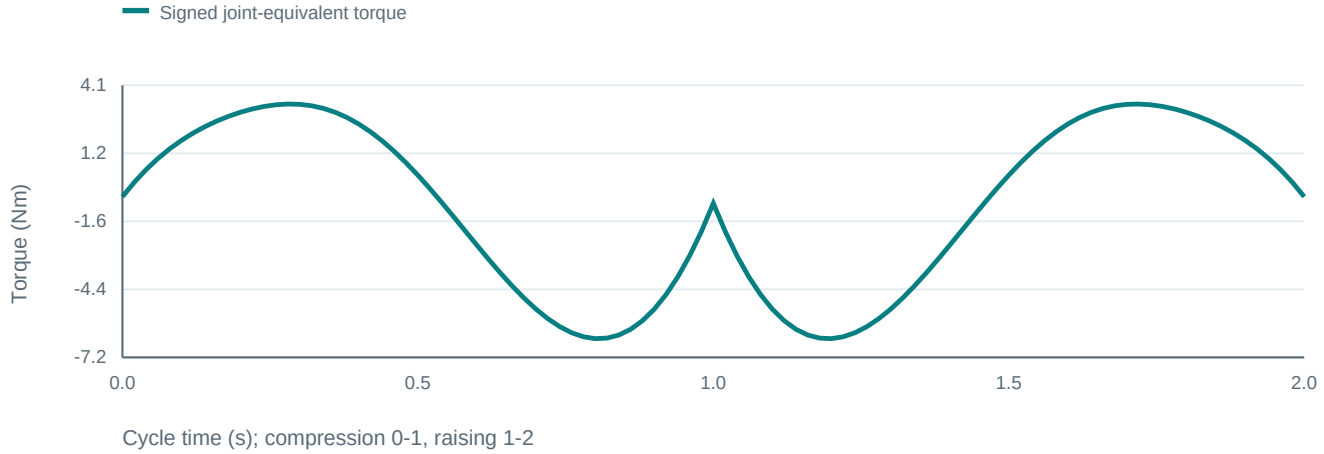
Red dashed spans and arrows show actual point-force directions used in the model. Pulley examples use ideal payout/tangent forces, not an added spring at the contact patch. All nine held-pose force-site virtual-work checks pass. Shapes are planar geometry, not a manufacturing drawing.

MOTION / BOTH DIRECTIONS

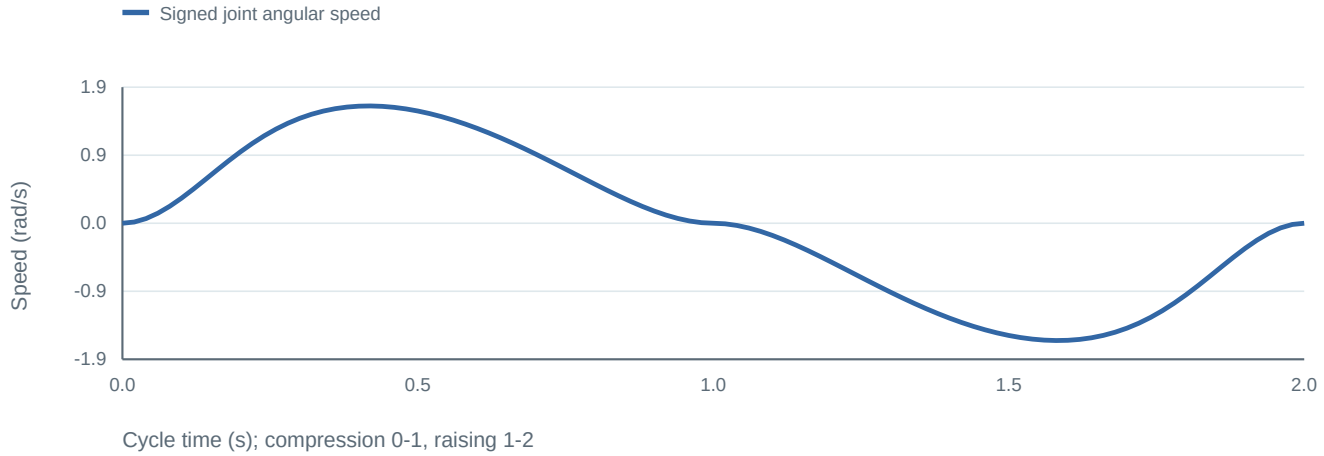
One second each way

C held fixed: compression lowers the chassis; reverse extension raises it.

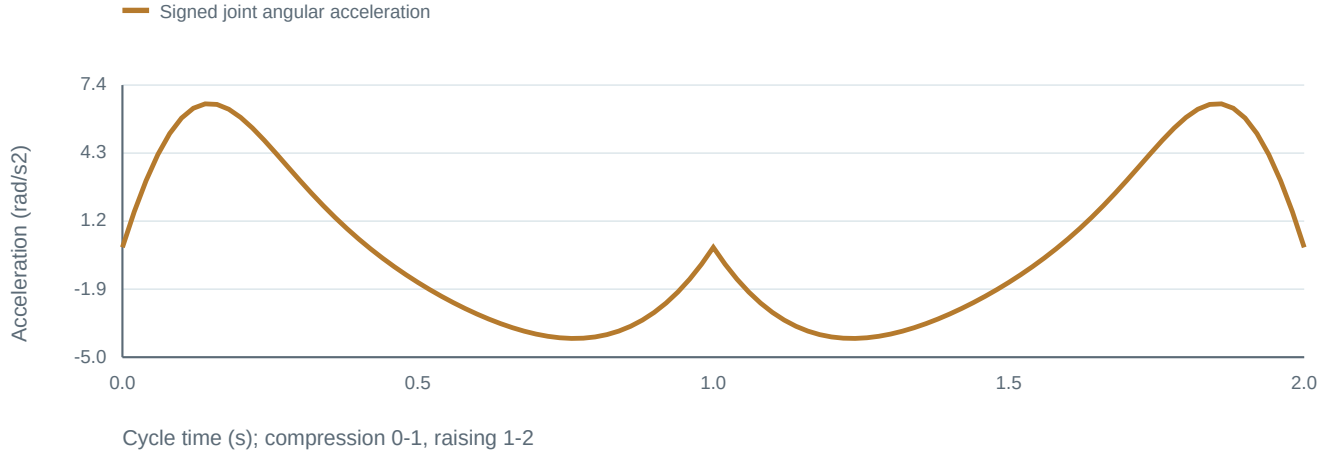
Signed joint-equivalent torque



Signed joint angular speed



Signed joint angular acceleration



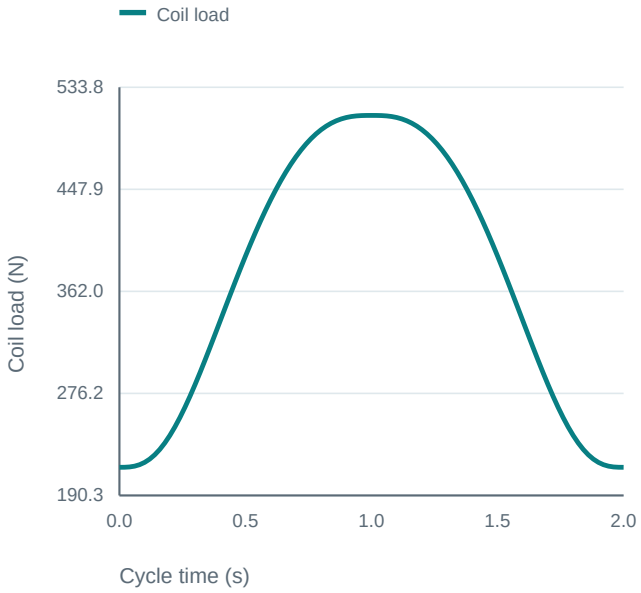
Full 2 s undamped out-and-back cycle; no dwell. Reported extrema/RMS use 201 points per direction; plots use a retained 101-point complete cycle. Reduction N=1 and eta=1.

SPRING / MOTOR SIZING

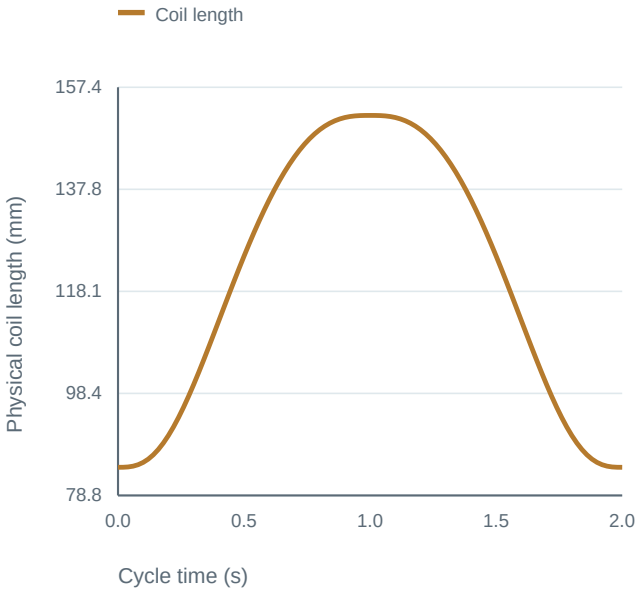
A spring is a torque profile

Selected hip bellcrank: real anchor span and passive Hooke tension; zero damping.

Load over both directions



Actual coil travel



Rate 4.369808 N/mm; installed preload 213.948 N; free length 35.253 mm; stroke 67.784 mm; peak load 510.150 N. Arm 80 mm at 135 deg; chassis mount (120, 50) mm. These mounts are illustrative, not packaging-approved.

Time each direction s	Peak Nm	Time RMS Nm	Peak speed rad/s	Peak accel rad/s2
0.5	23.939	13.887	3.210	26.292
1.0	6.447	3.641	1.605	6.573
2.0	2.111	1.135	0.803	1.643

Motor conversion is a declared assumption

For joint coordinate factor a (hip a=1; relative knee a=-qab): omega_motor=N*a*qdot; alpha_motor=N*a*qddot; tau_motor=tau_q/(a*N*eta). An illustrative hip N=10, eta=0.85 gives 0.759 Nm peak, 0.428 Nm RMS and 16.052 rad/s peak speed. That is 153.3 rpm and 65.731 rad/s2. This calculation omits unknown reflected motor/gear inertia and is not a purchased motor rating.

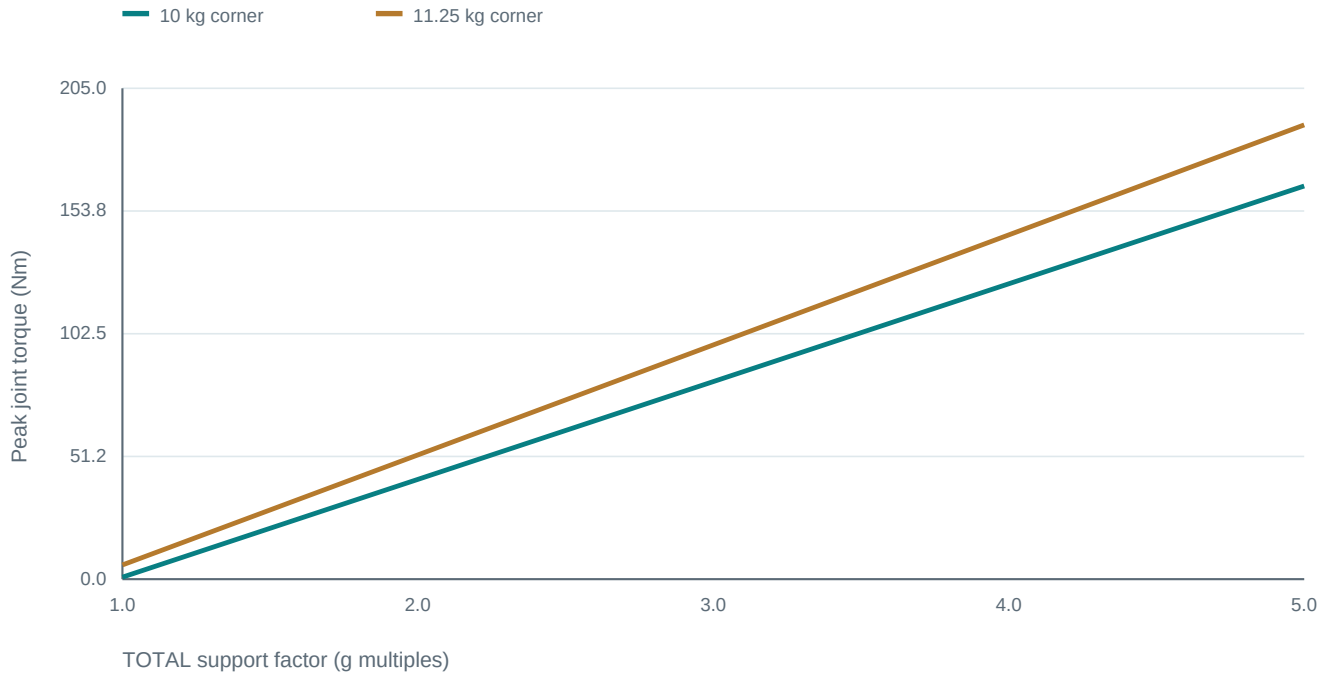
A symmetric undamped return preserves torque magnitude, reverses speed, and changes mechanical power sign. Efficiency is a conservative /eta torque convention; electrical regeneration and heating are unresolved.

LOAD ENVELOPE / HELD GEOMETRY

3g and 4g mean TOTAL support

Requested 4g net upward acceleration instead means 5g support. Keep both interpretations explicit.

Equivalent inertial stress; 1g-fitted spring unchanged



Total factor	10 kg support N	10 kg peak Nm	11.25 kg support N	11.25 kg peak Nm
1.0	98.10	0.838	110.36	5.943
3.0	294.30	82.515	331.09	97.829
3.5	343.35	102.934	386.27	120.800
4.0	392.40	123.353	441.45	143.772
5.0	490.50	164.191	551.81	189.715

This is a held-pose stress sensitivity: add equivalent downward inertial load (factor-1)*corner_mass*g at chassis A. Ordinary gravity appears once; no second multiplication of the one-second trajectory inertia occurs. The spring remains fitted for the nominal case.

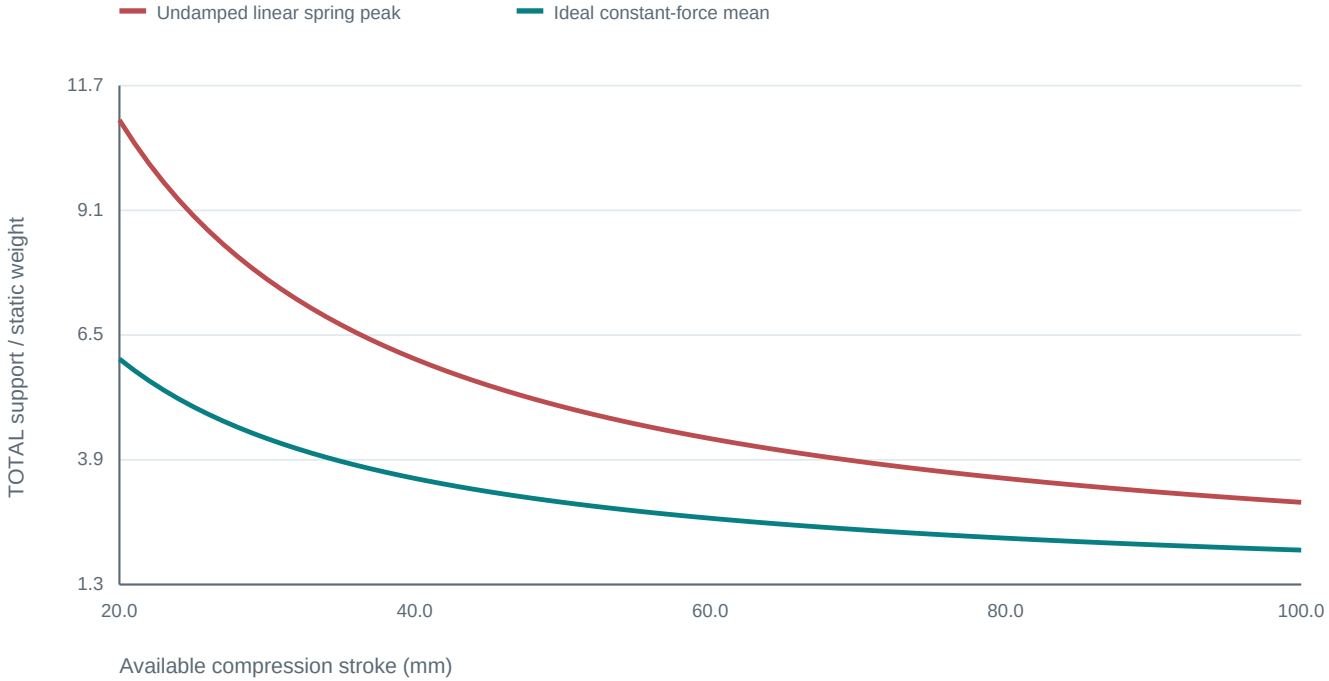
Zero damping stores and returns energy. These stress values do not integrate an impact, predict peak compression, select a shock waveform or establish dissipation. Contact sharing and remaining travel can dominate the actual response.

IMPACT / WHY STROKE MATTERS

A g limit needs an event

Analytical energy illustration; not a prediction of the generalized nonlinear spring.

Equivalent 100 mm drop; initial spring preload mg



Available stroke	Constant-force mean	Linear-spring peak
20 mm	6g	11g
50 mm	3g	5g
66.7 mm	about 2.5g	about 4g

```
h = v^2/(2g)
n_mean = 1 + h/delta
n_peak = 1 + 2h/delta
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For h=100 mm, v=1.401 m/s. A geometric 20 mm end buffer is not a declaration of compliant impact travel. Pick a ride pose and evaluate the real spring-energy rise and compression reserve.

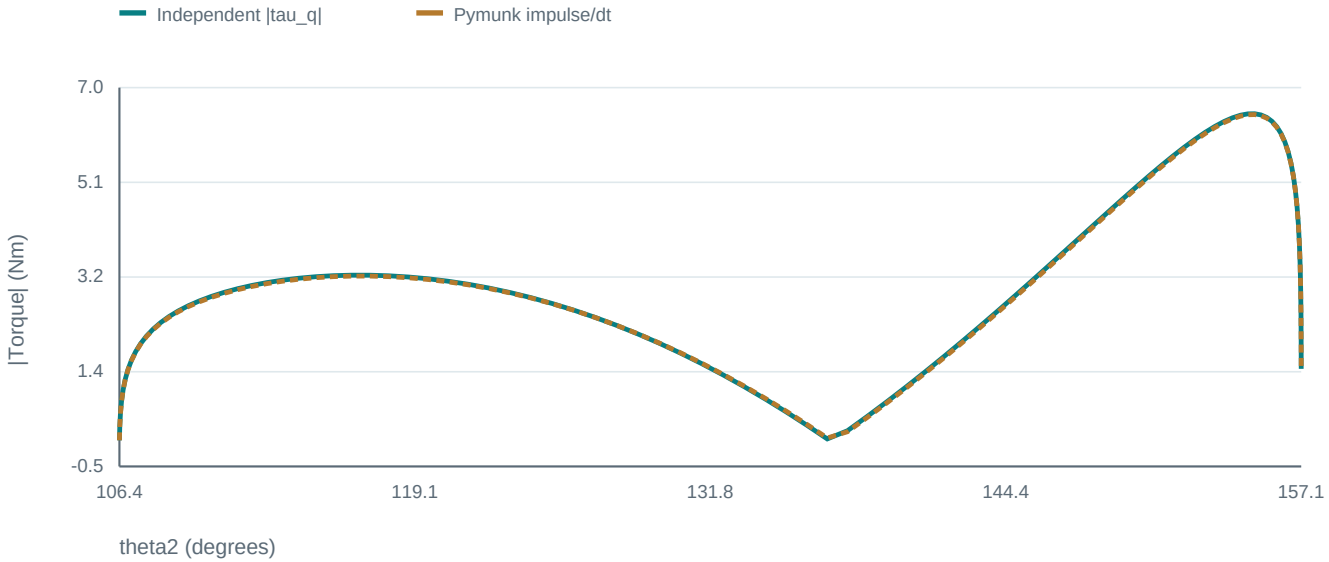
Unequal support sensitivity: at 45 kg whole-robot mass and only two supporting legs, equal sharing would be 22.5 kg per leg: 220.73 N at 1g, 662.18 N at 3g, 882.90 N at 4g. This is a load-sharing assumption, not an integrated two-contact scenario or a proven maximum.

VERIFICATION / NUMERICAL SCOPE

Actual engine, independent math

Constraint stepping observes motor impulse; analytical torque is used only for comparison.

Hip bellcrank, one-second compression, fine step



Topology	2 ms RMS error Nm	1 ms RMS error Nm	1 ms peak error Nm	Check
Legacy tip	0.1532	0.0767	0.1396	PASS
Hip pulley	0.0174	0.0087	0.0149	PASS
Knee pulley	0.1068	0.0534	0.0888	PASS
Direct scissor	0.0143	0.0072	0.0149	PASS
Hip bellcrank	0.0338	0.0169	0.0287	PASS
Knee bellcrank	0.0952	0.0477	0.0952	PASS
Chassis direct	0.0143	0.0071	0.0121	PASS
Knee capture	0.1068	0.0534	0.0888	PASS
Gravity balance	0.0522	0.0261	0.0384	PASS

Original solver: 3,853 kinematics + 16,044 statics + 29,863 global-peak assertions. New analytical suite: 13 tests; physical suite: 12 tests, including old Andy geometry/law cross-checks. All nine full dynamic checks, selected reverse motion and held-site virtual work pass.

Earlier source-app verification (development evidence): 85 JavaScript and 34 existing Python tests; full/embedded Edge checks, both directions and nine held checks pass. Source-app launcher lifecycles passed; this does not require a new Motion Lab integration.

Pymunk motor impulse is a magnitude: torque sign is independent analytical telemetry. No pose resets occur after initialization. Canonical robot physics is NOT APPLICABLE to this isolated member app; there is no hardware validation, soil, tire rolling or 3D collision proof.

REPRODUCTION / PROVENANCE

Keep the evidence with the model

Standalone PDF and offline static HTML are the delivery; no new tool integration is required.

Recorded input	Value / interpretation
Central geometry	L2/L3 180/340; ra/rb 106.5/60; rc/rd/re/rf 35; rw 200 mm
Angles / travel	alpha 4.375; beta 13.5; phi0 0; theta2 100-159.810888 deg; reserves20+20 mm
Physical mass allocation	m1 8.30; m2/m3 .35/.35; m5 .10; m6 .90 kg; total 10.0
Upper mass sensitivity	m1 9.55; same moving assembly; total 11.25 kg
Fixture / trajectory	C fully anchored; pitch held; 1 s quintic each direction; damping 0
Spring / transmission	Hip bellcrank / direct / extension; arm80 mm at135 deg; mount(120,50) mm
Actual engine check	Pymunk Space.step; 800 iterations; dt 2 and 1 ms; ideal pivots/gears
Motor reporting	Hip coordinate; N=1 eta=1 primary values; conditional N10/.85 example
Verified app environment	Windows11 Home26200; Python3.13.12; Node22.23.3; Pymunk7.3.0
Numerics / report pins	NumPy2.5.3; SciPy1.18.1; npm10.9.9; ReportLab4.4.9; pypdf6.10.0

Regenerate and recompute

From the app root (optional dependencies: requirements-report.txt):

```
python scripts/build_design_report.py
node --test tests/design-study.test.js
python -m unittest -v test_generalized_leg
python scripts/verify_generalized_leg.py
```

PDF generation uses ReportLab and pypdf. Numerical recomputation uses the documented app environment. The report data records exact configurations, scenario policies, profiles, source SHA256 fingerprints and base revision 6c74cfd.

Primary sources and repository map

Jonathan immutable **leg-solver/v1-kinematics.html**, **v2-quasi-static.html** and **v2-model.md**; Andy **docs/link-sizing.md**, **generalized-leg-math.md**, **generalized-leg-study.md**, **generalized-leg-physics.md** and **impact-load-envelope.md**. Background: Modern Robotics statics; Pymunk constraints; maxon RMS sizing; SDP/SI belts. Mechanism-specific results are computed repository evidence.

Unresolved hardware inputs

Measured masses/inertias and spring preload; tooth pitch/wrap/tension; contact friction; coil force/solid-height/fatigue limits; 3D shafts/chassis/clearance; reflected drivetrain inertia; motor efficiency, thermal duty and regeneration; measured impact event and contact sharing. The historical dual-rope design remains IMPOSSIBLE / rejected and is excluded.